

NWIS

DAILY DRILLING REPORT

Doc ID
W005-DDR-26

Well: Alpha-05 | Well ID: W005 | Field: Field_Alpha | Date: 24-01-2025

PRIMARY ISSUE CATEGORY: TORQUE SPIKE

1. REPORT SUMMARY

Report No.	26	Date	24-01-2025
Start Depth	2725 m	End Depth	2755 m
Progress (24 hr)	30 m	Formation	Formation_W2 (lower, stringer)
Rig	Rig Sagar-5	Mud Type	WBM - Polymer (+2% ester lubricant post-event)

2. MUD PROPERTIES

MW (SG)	PV (cP)	YP (lb/100ft2)	Gel 10s/10m	pH	Solids (%)
1.24	28	19	8/20	9.4	6.0

3. DRILLING PARAMETERS & TORQUE TREND

Depth (m)	ROP (m/hr)	WOB (klb)	RPM (actual)	MW (SG)	Torque (kft-lb)
2725	16	14	100	1.24	10.2
2732	14	14	95-135 (erratic)	1.24	14.8
2738	9	13	60-140 (erratic)	1.24	19.6
2740	5	10 (reduced)	70	1.24	22.0 (peak)
2745	8	10	72	1.24	12.5
2755	12	13	98	1.24 + lube	11.4

4. TORQUE & DRAG MODEL: ACTUAL VS. MODELED

Depth (m)	Modeled Torque (kft-lb)	Actual Torque (kft-lb)	Deviation
2700	9.8	10.1	+3%
2725	10.0	10.2	+2%
2738	10.5	19.6	+87% (stringer/dogleg)
2740	10.6	22.0	+108% (peak)
2755	10.8	11.4	+6% (post-treatment)

5. OPERATIONS TIMELINE (24 HR)

Time	Duration (hr)	Activity
00:00-04:00	4.0	Drilling ahead 2725-2735 m; torque near baseline

04:00-04:20	0.33	Torque spike 10 -> 22 kft-lb over 3 minutes at 2740 m
04:20-05:00	0.67	WOB reduced 30%, RPM reduced to 70; stick-slip breaking
05:00-07:30	2.5	Short wiper trip through build section; lubricant added to mud
07:30-08:00	0.5	Back to bottom; resumed drilling at reduced ROP
08:00-24:00	16.0	Drilling ahead 2740-2755 m; torque stable at 11-13 kft-lb

6. EVENT / INCIDENT DETAIL

At 2740 m, surface torque spiked from a 10 kft-lb baseline to 22 kft-lb within 3 minutes while drilling a hard calcareous stringer within a dogleg cluster, with RPM fluctuating erratically between 60-140.

Action Taken:

1. WOB reduced by 30% (from 14 klb to 10 klb) to reduce bit aggressiveness.
2. RPM reduced from 100 to 70 to help break the stick-slip cycle.
3. Real-time torque-and-drag model rerun and compared against actual surface torque.
4. Short wiper trip performed back through the build section.
5. 2% ester-based lubricant additive added to the active mud system.

Result: Torque stabilized at 11-13 kft-lb baseline after the wiper trip and lubricant treatment. ROP through the remaining stringer was reduced ~35% as a precaution.

NPT Incurred: 5.5 hours

Engineer Remark: Actual torque deviated from the pre-well model by over 100% at the peak; recommend updating the block friction factor assumption from 0.25 to 0.32 for this stringer/dogleg interval on future wells.

7. PERSONNEL ON SITE

Company Man	A. Mehta	Drilling Supervisor	D. Bhatt
Mud Engineer	V. Rao	Toolpusher	M. Das

This Daily Drilling Report is a synthetically generated historical record produced for the NWIS prototype (W005). Not associated with an actual drilling operation.